Tool Selection and Clarifications - Raw UCI MLOps Project

This document explains the tool stack used for the raw-only UCI Heart Disease MLOps assignment.

Category	Tool	Justification
Programming	Python	Strong ecosystem for data processing, ML, APIs, testing, and automation.
Data processing	pandas, NumPy	Used to parse raw 76-attribute UCI files, clean missing values, and create model-ready tables.
ML framework	scikit-learn	Provides pipelines, ColumnTransformer, GridSearchCV, and deployment-friendly estimators.
Tracking	MLflow	Logs experiments, metrics, plots, and model artifacts for reproducibility.
API	FastAPI	Modern, validated, async-ready API framework with automatic Swagger documentation.
Testing	pytest	Simple automated tests for data cleaning, preprocessing, training, and API endpoints.
Containerization	Docker	Reproducible runtime containing API, dependencies, raw-only data acquisition, and model training.
CI/CD	GitHub Actions	Academic-friendly automated workflow for linting, testing, training validation, and Docker build.
Orchestration	Kubernetes/Minikube	Local proof of deployment, scaling concepts, and service exposure.
Monitoring	Logging, Prometheus	Request logging and `/metrics` demonstrate basic production observability.
Deployment	Render/Railway	Lower complexity and easier screenshot evidence than full AWS/GCP/Azure setup.
Version control	Git/GitHub	Required for reproducible history, collaboration, submission evidence, and CI/CD integration.

Raw-only clarification: The processed UCI files are not used for model training. The acquisition script extracts the selected 14 variables directly from the four original raw 76-attribute source files.